Finite-depth Quasi-Phase Matching (QPM) and Converter Efficiency

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Finite lattice depth leaves each atom distributed over a finite region of a lattice site. The resulting phase spread reduces the coherent overlap between the atomic modes used by the optical and microwave conversion paths. This note first estimates that overlap, Q, and then inserts it into an extension of the transducer model as in https://github.com/AVKGitHub1/transduction_calcs.

1 What the finite-depth QPM factor means

Let u_j and v_j be the normalized collective coupling profiles of the optical/blue and UV/microwave paths, including their phases. Define

$$Q = \left| \sum_j u_j^* v_j \right|, \qquad \sum_j |u_j|^2 = \sum_j |v_j|^2 = 1, \qquad 0 \le Q \le 1. \tag{1}$$

For matched, uniform coupling envelopes, this becomes $Q = |\langle e^{i\Delta\mathbf{k}\cdot\mathbf{r}} \rangle|$, where, for the chosen conversion loop, $\Delta\mathbf{k} = \mathbf{k}_{780} + \mathbf{k}_{480} - \mathbf{k}_{\mathrm{UV}} - \mathbf{k}_{\mu}$. Thus Q is an *amplitude overlap*; Q^2 is the squared projection onto the useful collective mode. Neither quantity alone is the converter's success probability.

1.1 Localization within one site.

An ideal retroreflected lattice has period $d = \lambda_{\mathrm{L}}/2$ and reciprocal wavevector $K = 2k_{\mathrm{L}}$, with $k_{\mathrm{L}} = 2\pi/\lambda_{\mathrm{L}}$. For a well of depth U, the deep-lattice harmonic approximation gives [1]

$$V(x) = U \sin^2(k_{\mathrm{L}} x) \simeq U k_{\mathrm{L}}^2 x^2, \quad E_{\mathrm{R}} = \frac{\hbar^2 k_{\mathrm{L}}^2}{2m}, \quad \omega_t = \frac{2\sqrt{U E_{\mathrm{R}}}}{\hbar}. \tag{2}$$

The thermal position variance and the nth density Fourier harmonic are then

$$\begin{aligned} \sigma_x^2 &= \frac{\hbar}{2m\omega_t} \coth\!\left(\frac{\hbar\omega_t}{2k_{\mathrm{B}}T}\right), \\ b_n &= \left|\left\langle e^{inKx} \right\rangle\right| = e^{-(nK)^2\sigma_x^2/2} = \exp\left[-n^2 \sqrt{\frac{E_{\mathrm{R}}}{U}} \coth\!\left(\frac{\sqrt{U E_{\mathrm{R}}}}{k_{\mathrm{B}}T}\right)\right]. \end{aligned} \tag{3}$$

At $T = 0$, the hyperbolic cotangent tends to one: zero-point motion still gives $b_n < 1$ for any finite U and nonzero n. At fixed temperature, deeper trapping increases b_n toward one. The approximation requires localization near the well minimum; appreciable occupation near the barrier calls for actual band wavefunctions.

1.2 From one axis to the total overlap.

If $\Delta\mathbf{k} = \sum_i n_i \mathbf{K}_i$ exactly and the confined directions are independent, the factors multiply:

$$Q = \prod_i b_{n_i,i}; \qquad Q = b_{2,x} b_{2,y} \quad \text{for two second-order QPM axes.} \tag{4}$$

The last expression is the working geometry used below: co-propagating 780 and 480/481 nm fields, perpendicular UV, and suitable lattice axes. It is a *conditional geometry*, not fixed by the nominal wavelengths alone. A residual mismatch $\delta\mathbf{k}$ also produces phase cancellation across occupied site centres, through $|\langle e^{i\delta\mathbf{k}\cdot\mathbf{R}}\rangle|$. Beam angles, exact wavelengths, and mode envelopes must therefore be checked separately.

2 Analytical converter model with the orthogonal modes

2.1 Problem definition and collective-mode construction.

This is a derived spatial-mode extension of the supplied converter equations [3], in the atomic transduction setting of Ref. [2]. We seek the coherent conversion amplitude from an optical input to the microwave output when the two atomic coupling profiles have the overlap Q defined in Eq. (1). Set the notebook's $\epsilon = 1$ throughout.

Atomic variables and assumptions. Let $\hat{e}_j^\dagger$ and $\hat{r}_j^\dagger$ create an optical and a Rydberg excitation, respectively, on atom j relative to the prepared reference state. In the weak-excitation limit these operators are approximately bosonic:

$$[\hat{e}_j, \hat{e}_\ell^\dagger] \simeq [\hat{r}_j, \hat{r}_\ell^\dagger] \simeq \delta_{j\ell}, \qquad [\hat{e}_j, \hat{r}_\ell^\dagger] \simeq 0. \tag{5}$$

Assume frozen atomic positions during conversion, a uniform blue-control magnitude, and identical atomic detunings and independent decay rates across the ensemble. UV dressing is already included in the effective microwave coupling and its spatial profile, as in the calculator. There is no additional UV mode to solve for here. After absorbing the blue field's spatial phase into $\hat{r}_j$, its matrix element Ω can be chosen real and uniform. With the sign convention used below,

$$\frac{\hat{H}_B}{\hbar} = -\Omega \sum_j (\hat{e}_j^\dagger \hat{r}_j + \hat{r}_j^\dagger \hat{e}_j). \tag{6}$$

The two coupling profiles. In this phase convention, let $\mathbf{u}$ describe the collective optical excitation selected by the optical cavity, and $\mathbf{v}$ the Rydberg excitation selected by the UV-dressed microwave interaction. These are the profiles introduced in section 1:

$$\mathbf{u}^\dagger\mathbf{u} = \mathbf{v}^\dagger\mathbf{v} = 1, \qquad \mathcal{Q} = \mathbf{u}^\dagger\mathbf{v} = \sum_j u_j^* v_j, \qquad Q = |\mathcal{Q}|. \tag{7}$$

A global phase of $\mathbf{v}$ can be absorbed into the microwave cavity operator, so we henceforth choose $\mathbf{u}^\dagger\mathbf{v} = Q \geq 0$. Writing $s = \sqrt{1-Q^2}$, the microwave profile has the exact decomposition

$$\boxed{\mathbf{v} = Q\mathbf{u} + s\mathbf{w},} \qquad \mathbf{w} = \frac{\mathbf{v} - Q\mathbf{u}}{s}, \qquad \mathbf{u}^\dagger\mathbf{w} = 0, \quad \mathbf{w}^\dagger\mathbf{w} = 1. \tag{8}$$

The expression for $\mathbf{w}$ applies when $Q < 1$. At $Q = 1$, choose any normalized orthogonal profile: its coupling to the microwave cavity vanishes because $s = 0$.

Collective operators and closure. Define four atomic operators, using uppercase letters to distinguish the operators from the coherent amplitudes used later:

$$\begin{aligned} \hat{P}^\dagger &= \sum_j u_j \hat{e}_j^\dagger, \quad \hat{R}^\dagger = \sum_j u_j \hat{r}_j^\dagger, \\ \hat{P}_\perp^\dagger &= \sum_j w_j \hat{e}_j^\dagger, \quad \hat{R}_\perp^\dagger = \sum_j w_j \hat{r}_j^\dagger. \end{aligned} \tag{9}$$

To the same weak-excitation approximation, orthonormality gives unit commutators within each mode; distinct profiles commute. The optical cavity couples only to $\hat{P}$. Uniform blue control couples $\hat{P}$ to $\hat{R}$ and $\hat{P}_\perp$ to $\hat{R}_\perp$. The microwave cavity couples to $Q\hat{R} + s\hat{R}_\perp$. The two cavity profiles span at most a two-dimensional atomic subspace, preserved by the uniform control, detunings, and damping. Hence the coherently driven response closes on these four atomic modes plus the two cavity modes. All other orthogonal atomic modes have zero mean drive and are taken initially unexcited.

Hamiltonian and conventions. Let $\hat{a}$ and $\hat{b}$ denote the optical and microwave cavity operators. Retain $g_o = G/\sqrt{2}$, $g_m = G_\mu/\sqrt{2}$, and the blue-control matrix element Ω from the notebook, where G and G_μ are the calculator's G_{opt} and G_{mm}. In a rotating frame with zero optical-cavity detuning, the Hamiltonian restricted to the driven subspace is

$$\frac{\hat{H}_0}{\hbar} = -\Delta_\mu \hat{b}^\dagger \hat{b} - \Delta_e(\hat{P}^\dagger \hat{P} + \hat{P}_\perp^\dagger \hat{P}_\perp) - \Delta_r(\hat{R}^\dagger \hat{R} + \hat{R}_\perp^\dagger \hat{R}_\perp), \tag{10}$$

$$\frac{\hat{H}_{\mathrm{int}}}{\hbar} = -\Big[g_o \hat{a}^\dagger \hat{P} + \Omega(\hat{P}^\dagger \hat{R} + \hat{P}_\perp^\dagger \hat{R}_\perp) + g_m \hat{b}^\dagger (Q\hat{R} + s\hat{R}_\perp) + \mathrm{h.c.}\Big]. \tag{11}$$

The minus signs fix the detuning and coupling-phase convention to match Eq. (14): use $\dot{\hat{c}} = (i/\hbar)[\hat{H}, \hat{c}]$ and Fourier amplitudes proportional to $e^{+i\omega t}$. In this convention an optical drive with photon-flux amplitude $a_{\mathrm{in}}(t)$ is represented by

$$\frac{\hat{H}_{\mathrm{d}}}{\hbar} = -\sqrt{\kappa_{o,\mathrm{ex}}}\,\big[\hat{a}^\dagger a_{\mathrm{in}}(t) + a_{\mathrm{in}}^*(t)\hat{a}\big]. \tag{12}$$

Here $|a_{\mathrm{in}}|^2$ is incident photon flux. Define $a = \langle\hat{a}\rangle$, $b = \langle\hat{b}\rangle$, $p = \langle\hat{P}\rangle$, $r = \langle\hat{R}\rangle$, $p_\perp = \langle\hat{P}_\perp\rangle$, and $r_\perp = \langle\hat{R}_\perp\rangle$. With no incident microwave field, the useful output has amplitude $b_{\mathrm{out}} = \sqrt{\kappa_{m,\mathrm{ex}}}\,b$ up to an overall port phase, so $S_{mo} = b_{\mathrm{out}}/a_{\mathrm{in}}$. Reservoir noise has zero mean in this response calculation; the transfer efficiency does not characterize added noise.

2.2 Response functions and linewidth conventions.

Including damping, the above Fourier and detuning conventions give

$$\begin{aligned} D_a &= \kappa_o/2 + i\omega, & D_e &= \Gamma/2 + i(\omega - \Delta_e), \\ D_r &= \gamma_r/2 + i(\omega - \Delta_r), & D_b &= (\kappa_m + \gamma_r)/2 + i(\omega - \Delta_\mu). \end{aligned} \tag{13}$$

Here κ_o, κ_m are total cavity linewidths; Γ, γ_r set the atomic coherence damping as shown. The extra γ_r in D_b is inherited from the supplied model, not generated by QPM. The desired port rates are $\kappa_{o,\mathrm{ex}}$ and $\kappa_{m,\mathrm{ex}}$.

2.3 Six coupled equations.

The coherent amplitudes defined above obey, with the chosen input phase,

$$\begin{aligned} D_a a - i g_o p &= i\sqrt{\kappa_{o,\mathrm{ex}}}\,a_{\mathrm{in}}, \\ D_e p - i g_o a - i\Omega r &= 0, \\ D_r r - i\Omega p - i g_m Q b &= 0, \\ D_b b - i g_m (Q r + \sqrt{1-Q^2}\,r_\perp) &= 0, \\ D_r r_\perp - i\Omega p_\perp - i g_m \sqrt{1-Q^2}\,b &= 0, \\ D_e p_\perp - i\Omega r_\perp &= 0. \end{aligned} \tag{14}$$

The microwave mode couples to $Qr + \sqrt{1-Q^2}\,r_\perp$. Its total collective coupling remains g_m; the useful component is Qg_m.

2.4 Eliminating the orthogonal pair.

Writing $F = D_e D_r + \Omega^2$, the last two equations give

$$r_\perp = \frac{i g_m \sqrt{1-Q^2}\, D_e}{F}\, b, \qquad \boxed{\Sigma_\perp(\omega) = \frac{g_m^2 (1-Q^2) D_e}{F}}. \tag{15}$$

The microwave equation therefore contains $D_b + \Sigma_\perp$. Its real part adds damping and its imaginary part changes the dispersive response. Simply replacing G_μ by QG_μ drops this response. It must not be added to the useful output-port rate.

2.5 Closed transfer function.

Define $Y = D_a F + g_o^2 D_r$ and $Z = D_b F + g_m^2 D_e$. Solving the remaining equations gives, up to an overall output phase,

$$\boxed{S_{mo}(\omega) = \frac{\sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, g_o g_m \Omega Q F}{YZ + Q^2 g_o^2 g_m^2 \Omega^2}, \qquad \eta(\omega) = |S_{mo}(\omega)|^2.} \tag{16}$$

Q changes both the numerator and the resonant response in the denominator. This is why the original efficiency cannot generally be scaled by Q^2. At $Q = 1$, the orthogonal pair decouples and the original four-mode model is recovered; at $Q = 0$, coherent conversion vanishes.

3 On-resonance efficiency and the numerical value of Q

At $\omega = \Delta_e = \Delta_r = \Delta_\mu = 0$, introduce

$$C_o = \frac{2G^2}{\kappa_o \Gamma}, \qquad C_m = \frac{2G_\mu^2}{(\kappa_m + \gamma_r)\gamma_r}, \qquad C_B = \frac{4\Omega^2}{\Gamma \gamma_r}. \tag{17}$$

Let $\eta_o = \kappa_{o,\mathrm{ex}}/\kappa_o$ and $\eta_m = \kappa_{m,\mathrm{ex}}/\kappa_m$. Equation (16) becomes

$$\boxed{\eta(0) = \eta_o \eta_m \frac{\kappa_m}{\kappa_m + \gamma_r} \frac{4Q^2 C_o C_m C_B (1+C_B)^2}{\left[(1+C_B+C_o)(1+C_B+C_m) + Q^2 C_o C_m C_B\right]^2}.} \tag{18}$$

The port fractions multiply this expression at fixed total linewidths. If a change in mirror or microwave-port coupling changes a total linewidth, that linewidth and the cooperativities must also be updated. Additional propagation or detection losses multiply the port-to-port result when defining an end-to-end efficiency.

3.1 Lattice example.

Use ^{85}Rb, $\lambda_\mathrm{L} = 1188\,\mathrm{nm}$, waist $w = 200\,\mu\mathrm{m}$, 100% retroreflection, and the specified calibration of 3 W per beam per axis to $U/k_\mathrm{B} = 15\,\mu\mathrm{K}$. This is an adopted depth calibration, not a new polarizability calculation. The recoil energy is $E_\mathrm{R}/k_\mathrm{B} = 0.0799004\,\mu\mathrm{K}$. For $T = 0$ and two equal, exactly matched second-order axes,

$$b_2 = e^{-4\sqrt{E_\mathrm{R}/U}} = 0.746816, \qquad \boxed{Q = b_2^2 = 0.557734}, \qquad Q^2 = 0.311067. \tag{19}$$

The $T = 0$ value is the harmonic lowest-band approximation and includes zero-point motion. It is a static overlap input to the model; motion-resolved sidebands are not calculated here.

3.2 Converter example.

Use the screenshot's parameters, in the calculator's MHz convention (angular rates divided by 2π):

$$G = 15.8474565, \quad G_\mu = 2.09170145, \quad \Omega = 5.52497339,$$
$$\kappa_o = 10, \quad \kappa_m = 5, \quad \Gamma = 6, \quad \gamma_r = 0.1.$$

All detunings are zero. A common conversion of every rate to angular units leaves the efficiency unchanged. With $\eta_o = \eta_m = 1$:

Model	On-resonance efficiency
Original converter, $Q = 1$	82.0957%
Only $G_\mu \mapsto QG_\mu$; orthogonal pair omitted	51.5929%
Six modes with $Q = 0.557734$	**47.0824**%

Here $\Sigma_\perp(0) = 0.147394\,\mathrm{MHz}$, corresponding to an additional full linewidth $2\,\mathrm{Re}\,\Sigma_\perp = 0.294787\,\mathrm{MHz}$ in the same convention. For example, desired-port fractions of 0.98 at both cavities give 45.21798% at unchanged total linewidths, before any further path losses.

3.3 Scope and checks.

The closure assumes weak excitation, a frozen spatial distribution, uniform control magnitude and atomic linewidths, and one effective optical mode profile. Anharmonic motion, resolved motional sidebands, differential light shifts, and further independent standing-wave channels require an extended susceptibility model. The closed expression was checked against the six-equation solution and the $Q = 1$ limit in Mathematica. The efficiencies above are conditional model predictions for the stated geometry.

References

[1] I. Bloch, J. Dalibard and W. Zwerger, *Rev. Mod. Phys.* **80**, 885 (2008), Sec. II. doi:10.1103/RevModPhys.80.885.
[2] A. Kumar et al., *Nature* **615**, 614–619 (2023). doi:10.1038/s41586-023-05740-2.
[3] Supplied `Abhishek_Converter.nb`; accompanying `Abhishek_Converter_QPM.nb`. Calculator: AVKGitHub1/transduction_calcs, commit `30da8841b3c9`. The Q extension is derived here.